Behind-the-Meter Battery Site

as a Capacity Service for PJM Data Centers

(Phase 1)

Jiawei Qi

Advisor: Prof. Nathan Williams (CMU Africa)

Carnegie Mellon University

Independent Study

Contents

1. Executive Summary and Recommendation 1

2. Project Definition and Business Model 1

3. Market and Stakeholder Case 2

4. Assumptions, Sources and Analytical Boundary 4

5. Technical Feasibility 4

6. Financial Feasibility 5

7. Sensitivity Analysis and Conditions for Success 7

8. Environmental and Social Considerations 8

9. Regulatory and Contractual Requirements 9

10. Risk Analysis and Development Plan 10

11. Final Appraisal and Next Steps 11

References i

Appendix A. Recharge Between Consecutive Dispatch Days iii

Appendix B. Development Plan Detail iii

1. Executive Summary and Recommendation

A third-party developer builds, owns and operates a 50 MW / 500 MWh lithium iron phosphate (LFP) battery storage site behind the customer's meter, serving a new 50 MW co-location data-center operator's load in Loudoun County, Virginia, from 1 June 2029. The project responds to PJM's proposed Interim Resource Adequacy Service (IRAS), under which a new load of 50 MW or more that does not bring its own new capacity (BYONC) is reduced first in a capacity shortage. Registered as the load's new demand response, the battery supplies that capacity, with 45.0 MW accredited against the load's 46.16 MW obligation. It discharges only when PJM dispatches load management and is held full at all other times. In return, the customer pays a fixed availability fee per kW of load for 15 years, and the developer keeps the battery's PJM capacity revenue.

The market for this service is real. PJM's 2028/29 auction cleared 6,831 MW short, and the Dominion zone has already had 13 days and 82.2 hours of demand-response dispatch since January 2025. Since backup generators cannot routinely fill the gap, a co-location operator whose load is reduced first would break its firm-service promise to tenants. Our design meets this duty. With 10 hours of storage, it covers 15 of the 16 recorded dispatch days, and the 1.16 MW it cannot accredit is tested only in the first year and can be bought elsewhere in PJM. The base case is also financeable. Its $135.1M of capital cost (CAPEX) is funded by $55.7M of debt and $79.4M of equity, and selling the 30% tax credit returns $36.5M in the first year, so a minimum fee of $27.65/kW-month gives equity a 12% return and the project 8.68%, with a minimum debt service coverage ratio (DSCR) of 1.42x for lenders. Installed cost moves the fee more than any other input, from $20.22 to $35.62 across the National Renewable Energy Laboratory's (NREL) 2028 low and high projections.

The customer side is not yet shown. For 2029/30, no route with a known lower price gives the customer the same result, but once PJM's Reliability Backstop Procurement (RBP) has capacity in service, and if Virginia charges its cost to data centers without their own capacity, the customer could reach the same result for at most $15.58/kW-month, and $6.10 at the 2029/30 price cap. Matching that price would need CAPEX at 39% to 65% of the base. The battery's advantage is therefore concentrated before RBP capacity arrives, while the contract runs 15 years. Siting and approvals are workable, but untested. Loudoun permits storage by right in industrial districts, subject to setbacks, a 5-acre minimum and no historic resource within 1 mile, yet fire, noise and decommissioning still need independent study, and the Virginia State Corporation Commission's (SCC) storage permit has no decision deadline. That matters because financial close on 31 December 2027 leaves 17 months to build.

These findings leave five conditions for success:

- The Federal Energy Regulatory Commission (FERC) must accept IRAS with its BYONC rules
- PJM must accept the battery-backed site as a demand resource at the 90% rating
- A customer must sign a 15-year contract at or above $27.65
- The installed cost must stay at or below $135.1M with the tax-credit conditions met
- The battery must be registered and in service by 1 June 2029

The recommendation is therefore to study the project further, continuing on a limited budget to the go/no-go gate of 31 March 2027. Until then, the developer values the service from the customer's side and prices cheaper options, such as a 5-hour design ($12.91, covering 5 of 16 dispatch days), a lower installed cost or a term matched to the years before RBP capacity arrives, and commits to vendor, permitting and financing spend only if the gate is passed.

2. Project Definition and Business Model

2.1 Offer, Technology and Location

PJM's 2028/2029 capacity auction cleared 6,831 MW short of the reliability requirement [1], and PJM filed IRAS at FERC on 13 August 2026 [2, 3]. The project sells the load's BYONC as a service: the battery is registered as the load's new demand response, so the load keeps its status even if the

battery under-performs, and the developer pays any PJM charges. The battery sells no energy, is not outage backup and does not speed up the grid connection.

The battery is a 50 MW / 500 MWh LFP system [4]. Its power equals the load, because a load-reduction resource cannot cut more than the site draws, and its 10 hours cover every recorded Dominion-zone dispatch since 2017 except Winter Storm Elliott [5]. At PJM's 90% demand-resource rating for 2029/30 [6], it accredits 45.0 MW, 97.5% of the load's 46.16 MW obligation (50 MW × the 0.9232 Forecast Pool Requirement [7]). The customer arranges the remaining 1.16 MW, which PJM tests in the first delivery year only.

Loudoun County held 71 of the 131 data-center sites that Virginia's legislative audit commission mapped in 2024 [8, Table B-1] and lies in Dominion Energy Virginia's territory and PJM's Dominion zone. The battery sits on a representative new co-location campus there, behind Dominion's revenue meter at the campus switchgear, so it needs no PJM generator interconnection.

2.2 Customer and Key Actors

The customer rents firm capacity to its tenants at $160/kW-month or more [9], so a load that is reduced first puts its product at risk. The developer, which is also the equity sponsor, finances, builds, owns and operates the battery for a 12% equity return, so the customer avoids $135.1M of capital and stays out of PJM's markets. Around them, PJM runs the capacity market and FERC approves its rules, Dominion serves the load and reports the BYONC, a Curtailment Service Provider (CSP) registers the battery, the SCC, the Department of Environmental Quality (DEQ) and Loudoun County issue the permits, the integrator and the engineering, procurement and construction (EPC) contractor build the battery, and lenders and a tax-credit buyer finance it.

2.3 Value Creation and Capture

The service creates value for both the customer and the grid. The new load keeps firm service, and at the industry-average power usage effectiveness (PUE) of 1.54 [10] it holds 32.47 MW of IT capacity worth $62.3M a year in rent, while PJM gains 45.0 MW of new accredited capacity. The developer captures this value through the fee, $16.59M in 2029/30 at the $27.65/kW-month minimum, and through PJM capacity revenue of $5.55M. The developer bears construction, performance, PJM charges and the capacity price; the customer bears the fee, the 1.16 MW and the regulatory risk.

2.4 Development Decision

The decision evaluated is whether the developer should spend a limited budget to reach the go/no-go gate on 31 March 2027 and, if passed, proceed to financial close for service on 1 June 2029.

3. Market and Stakeholder Case

3.1 Customer Need

Under IRAS, a new load without BYONC or backstop credit is reduced even before PJM's Pre-Emergency Load Management, so a new campus without its own capacity cannot promise firm service. Such reductions are not rare: PJM's postings record 16 dispatch days in the Dominion zone since January 2017, 102.7 hours in all, of which 13 days and 82.2 hours, or 47.8 hours a year, fell after 1 January 2025. IRAS applies only in shortfall years, and BYONC obligations lapse if it is not triggered for three consecutive years.

Backup generators cannot fill the gap routinely, since emergency engines may run at most 50 hours a year outside emergencies and not for demand response [11], and for data-center air-permit applications from 1 July 2026 Virginia requires every generator set to meet Tier 4-equivalent emissions [12]. The IRAS credit of $1,593/MWh [13] exceeds the $142 of rent the load earns per MWh, so what the operator stands to lose is the firm-service promise itself, in an industry where 57% of operators say their last major outage cost over $100,000 and one in five over $1 million [14].

3.2 Demand, Market Size and Offtake

CBRE reports that Northern Virginia held 4.50 GW of data-center inventory in mid-2026, with 2.42 GW under construction and 0.2% vacancy, and Dominion forecasts its data-center demand to grow from 4.2 GW to 7.0 GW by 2030 [15]. Measured in capacity, if all 2.8 GW of that growth were new loads under IRAS, leaving the first tier would need 2.58 GW of new accredited capacity (2.8 GW × 0.9232), the upper bound of the addressable market. If Dominion's seven largest customers, which used 72% of year-to-date demand, are hyperscalers that arrange their own supply, co-location and other operators account for 28% of the growth, 0.78 GW, enough for 15 loads of 50 MW. The obtainable market is one campus, with one customer, one 50 MW load and one 15-year take-or-pay agreement, and no customer has been approached. Price is not the constraint, not demand.

3.3 Alternatives and Competition

Besides this project, the customer has six routes to the same result, its new load not reduced first from 1 June 2029. Table 3.1 compares them by the extra cost each adds per kW-month of load, since the customer's Dominion bill is the same on every route. RBP buys new capacity on 15-year contracts at an average of no more than $555/MW-day [16].

Table 3.1 Customer's Routes to Firm Service for a New 50 MW Load

Route	Service from 1 June 2029	Extra cost ($/kW-month)	Main drawback
This project	Yes, if on schedule	27.65	15-year take-or-pay
Build battery itself	Yes	27.65 by itself	$135.1M of customer capital
Buy another new capacity	Only if built in time	Not published	Virginia may require local capacity
Rely on RBP	Only RBP capacity is in service	6.10–15.58	Depends on FERC and Virginia
Build or contract on-site gas	Unlikely	Not estimated	24-month build; air permit
Register its own load	No	0, but power is cut	PJM charges for any shortfall
Accept IRAS reductions	No	0, but power is cut	Reduced first in every shortage

For 1 June 2029, no route with a known lower price gives the same result. The customer's own battery costs the same at the same financing, and off-site capacity has no published price; on-site gas costs 48–93% as much as the battery to build per accredited kilowatt [17], but its fuel, emission controls and air permit are not costed and a 24-month build makes 2029 unlikely. RBP is the one priced route below the fee, but it protects the customer only once its capacity is in service and if Virginia charges the customer for it. FERC has suspended RBP to 28 February 2027 pending a hearing, and RBP resources may arrive as late as 1 June 2032 [18]. Once in service, RBP caps what the customer will pay, since the fee is $1,010 per accredited MW-day against RBP's $555. The battery's advantage is therefore concentrated in the years before RBP capacity arrives, while a 15-year take-or-pay keeps the customer paying after it does. Cutting load costs no fee but leaves the customer with its outage loss, which the next stage estimates.

The suppliers behind these routes are the project's competitors. Enchanted Rock sells on-site gas generation to data centers as a service [19]; any new resource that wins RBP or sells BYONC bilaterally offers the same reliability to the same loads; and Dominion's statutory mandate to seek 16 GW of short-duration and 4 GW of long-duration storage by 2045 [20] competes for equipment and contractors rather than for this contract.

3.4 Stakeholders and Institutions

The rules that create this market are federal, so the decisive institutions are FERC, which decides IRAS and RBP, and PJM, which sets accreditation, registration and dispatch. Both decisions are contested. The market monitor and 22 other parties filed protests or limited protests against IRAS [21, 22], and PJM's governors want RBP costs to fall on data centers without their own capacity, with Virginia's retail rules to follow. Locally, Loudoun's Board of Supervisors and its residents decide land

use [23], and the customer must accept a third party inside its most protected system, here 500 MWh of cells beside the data halls.

3.5 Implications for Scale, Price, Revenue and Service Delivery

Scale is set by the load: each further 50 MW load needs another battery of this design. Price is bounded below by the developer's $27.65/kW-month and, once RBP capacity is in service, above by the customer's $6.10–$15.58, which only the 5-hour design ($12.91) falls within, near the top of that range. The fee is 75% of revenue, so the 15-year term is the hardest part to sell. Service delivery requires the BYONC to be reported before the auction it clears and each dispatch to be held for all hours [24].

4. Assumptions, Sources and Analytical Boundary

4.1 Analytical Boundary

The technical boundary runs from the battery's cells to the campus medium-voltage switchgear, as costed by NREL, while the customer's load, campus system and generators, Dominion's network and the customer's 1.16 MW lie outside it. The financial boundary is the project company, which holds one battery financed by project debt and sponsor equity and is judged by the equity investor at 12% and by lenders on coverage, so the customer's own bill, RBP charges and IRAS credits serve only as benchmarks. Interest during construction and financing fees, decommissioning, CSP fees, pre-construction development and network upgrades are left out, and each would raise the fee, while site rent and recharge demand charges fall on the customer by contract.

4.2 Units, Currency, Inflation and Timing

Values are nominal US dollars in millions; the fee is per kW of load per month, and capacity prices are per MW-day of unforced capacity. NREL's 2024 cost is taken at its 2028 mid projection and escalated at 2.5% a year to 2028; O&M and charging losses keep escalating, while the fee, capacity price, property tax, insurance and PJM charges are flat. Model years are PJM delivery years: Year 0 is 2028/29, and Years 1–15 run from 2029/30 to 2043/44, NREL's battery life, with year-end cash flows discounted to the end of Year 0.

4.3 Principal Assumptions and Sources

The inputs that move the result are the capacity obligation of 50 MW × 0.9232 = 46.16 MW; accreditation at 90% in 2029/30, falling to 75% by 2042/43 [25]; installed cost at NREL's 2028 mid projection, $135.1M; fixed O&M at 4% of installed cost; forced outages of 8.94% [26]; PJM charges of $3,185/MWh in a Performance Assessment Interval (PAI); the capacity price at the 2029/30 cap of $337.83/MW-day, held flat; a 30% tax credit sold at $0.90 [27, 28]; debt at 7.02% for 15 years, sized at 1.20x contracted and 2.0x merchant coverage [29, 30]; and a 12% equity return. The workbook's Inputs sheet lists all 73 inputs.

4.4 Sourced Values and Own Estimates

Of the 73 inputs, 41 are sourced as published, 1 is derived, 3 are points within published ranges, 6 are applied beyond their basis (such as 4-hour O&M and California outages), 3 are held constant by judgment (such as the capacity price), 13 are PJM preliminary ratings, 1 is held at the last published value, 4 are design choices (in-service date, term, flat fee and duration) and 1 is the solved fee.

4.5 Assumptions with the Greatest Effect

The installed cost, capacity price and tax credit move the fee most, while FERC's acceptance of IRAS and PJM's 90% rating decide whether the service is needed and how much it covers. Outside the model, the customer's alternatives set the price the fee must meet.

5. Technical Feasibility

5.1 Configuration

The system consists of LFP battery containers with modules, racks, battery management, thermal management and fire suppression, connected through bidirectional inverters and transformers to

the campus switchgear and run by SCADA controls. When PJM dispatches load management in the Dominion zone, the battery cuts the site's draw by up to 50 MW for all hours of the dispatch. It never exports and is held full between events rather than trading, because charging through the customer's meter would add to the customer's billed demand. Two other configurations were rejected. All-hours demand headroom does not work because Dominion bills the highest 30-minute demand of 12 months [31], and a front-of-meter battery would need its own PJM interconnection.

5.2 Scale and Output

Each quantity follows from a rule or a record. The 50 MW load is PJM's Large Load threshold, and the battery's power matches it. A larger battery cannot close the 1.16 MW gap, since it would need 46.16 ÷ 0.90 = 51.3 MW of reduction, more than the load. PJM tests the obligation in the first year only, after which the first-year installed capacity (ICAP) must be kept for 10 years, and BYONC may be a portfolio of new resources anywhere in PJM, so the customer can buy the gap elsewhere once. The 16 recorded dispatches lasted 1.75 to 9 hours, except Winter Storm Elliott at 14.5 hours [32]. Because each must be held for all hours, the longest normal event sets the duration at 10 hours and the energy at 500 MWh. On the same record, 5 hours covers 5 of the 16 days at a fee of $12.91, and 15 hours covers all 16 at $42.71, 54% above the base and outside NREL's 1–10 hour cost range; Elliott's extra 4.5 hours are paid as a PJM charge. Accredited capacity falls with PJM's ratings from 45.0 MW to 37.5 MW by 2042/43.

5.3 Performance and Reliability

With no published PJM battery outage rate, the model uses the largest measured fleet figure, 8.94% forced outages for 99 California batteries in 2022–2024; undelivered energy is charged at $3,185/MWh in a PAI and half that otherwise. NREL's 4% fixed O&M covers augmentation for about one cycle a day, while this battery would have been dispatched on 13 days since January 2025, so the 4% is more likely high than low.

5.4 Site, Infrastructure and Logistics

The battery occupies 3.0–8.0 acres on a lot that must meet Loudoun's rules for storage, namely a 5-acre minimum, 100-foot setbacks and no historic resource within 1 mile [33, §4.07.02], and a vendor layout is a first task. On the grid side, the campus switchgear must take a 50 MW bidirectional feed, the customer installs relays to Dominion's specification, and any demand above the contract permanently raises the contract; NREL's cost includes a substation and line that the battery does not need. On the supply side, a 2028 construction start needs at least 65% of its content free of foreign entities of concern (FEOC) to keep the tax credit [34].

5.5 Operating Constraints

Eleven of the 16 dispatch days fall in runs of two or three, so the battery must refill overnight with 18.2–32.5 MW above load (Table A.1). Without recharge it covers 10 days, PJM charges rise from $0.157M to $0.374M a year and the fee rises to $28.01, so the customer must keep up to 32.5 MW of recharge room or bear the shortfall. Because dispatches occur in every season, the battery is held full all year, with maintenance in March, April and November, months with no dispatch on record.

5.6 Inputs to Cost and Revenue

Energy and power capacity set CAPEX and, through it, O&M, insurance and property tax; recorded events set charging losses and PJM charges; and accredited capacity sets capacity revenue.

6. Financial Feasibility

6.1 Capital Cost, Operating Cost and Revenue

CAPEX builds up from the design quantities and NREL's 2024 unit costs, which include fire suppression, switchgear, permitting, contingency and developer overhead, at NREL's 2028 mid projection. Debt of $55.7M (41.2%) and equity of $79.4M fund it. Operating cost rises from $7.38M in 2029/30 to $9.67M by 2043/44 as O&M and charging losses escalate. The capacity price is the

2029/30 cap, held flat because the last three auctions cleared at their caps [35–37], and capacity revenue falls to $4.62M as PJM's rating falls to 75%. Table 6.1 gives the first-year figures.

Table 6.1 Base-Case Capital Cost, Operating Cost and Revenue

Item	Basis	Million USD
	Capital cost, 2028/29	
Energy component	500,000 kWh × $241/kWh (2024$)	+ 120.5
Power component	50,000 kW × $372/kW (2024$)	+ 18.6
Total CAPEX	(120.5 + 18.6) × 0.88 (NREL 2028 mid) × 1.025^4 ($2,702/kW)	= 135.1
	Operating cost, 2029/30	
Fixed O&M with augmentation	4% × $122.4M × 1.025^5	+ 5.540
Property tax and insurance	0.84% and 0.4% of CAPEX [38]	+ 1.675
Charging losses and PJM charges	89 MWh at $47.53/MWh [39]; 30.5 MWh × $3,185 + 37.8 MWh × $1,593	+ 0.162
Total operating cost	Sum ($148/kW-year)	= 7.377
	Revenue, 2029/30	
Availability fee	$27.65/kW-month × 50,000 kW × 12	+ 16.59
Capacity revenue	45.0 MW × $337.83/MW-day × 365	+ 5.55
Total revenue	Sum; capacity is 25.1%	= 22.14

6.2 Tax Credit and Depreciation

The 30% investment tax credit also needs prevailing wages and apprenticeships, and a project with early losses sells it at $0.90 per dollar [40]. Half the credit reduces the basis [41] for 5-year Modified Accelerated Cost Recovery System (MACRS) depreciation [42]; federal and Virginia taxes combine as below [43, 44], and losses offset at most 80% of later income [45]:

$$ITC\ cash\ =\ 30\%\ \times\ \$135.1M\ \times\ 0.90\ =\ \$36.5M$$
$$Depreciable\ basis\ =\ \$135.1M\ -\ 50\%\ \times\ \$40.5M\ =\ \$114.8M$$
$$Tax\ rate\ =\ 21\%\ +\ 6\%\ \times\ (1\ -\ 21\%)\ =\ 25.7\%$$

The credit returns 46% of the equity in 2029/30. The project pays its first income tax in 2034/35 and $11.2M in total.

6.3 Financing Structure

Lenders size debt on the lowest cash flow available for debt service (CFADS) at 1.20x for contracted and 2.0x for merchant revenue, at 7.02%, made up of the 4.77% ten-year Treasury yield plus a 1.75% margin, a 0.25% storage premium and 0.25% for uncontracted revenue above 10%. Goal Seek then solves the fee F that gives equity 12%:

$$DSCR\ target\ =\ 74.9\%\ \times\ 1.20\ +\ 25.1\%\ \times\ 2.0\ =\ 1.40x$$
$$Debt\ =\ PV\ at\ 7.02\%\ over\ 15\ years\ of\ (\$8.58M\ \div\ 1.40)\ =\ \$55.7M$$
$$Equity\ NPV\ =\ \Sigma\ (t\ =\ 0..15)\ ECF_t(F)\ \div\ 1.12\text{^}t\ =\ 0\ \rightarrow\ F\ =\ \$27.65/kW-month$$

6.4 Results and Interpretation

Of $321.4M of revenue over 15 years, 39.5% pays operating cost, 28.6% debt service at a level $6.12M a year, 3.5% income tax and 28.5% reaches equity, besides the tax credit. Table 6.2 gives the results.

Table 6.2 Base-Case Results

Metric	Value
Minimum availability fee	$27.65/kW-month
Equity IRR; project IRR (unlevered, after tax)	12.0%; 8.68%
DSCR: minimum (2043/44), average, target	1.42x, 2.00x, 1.40x
Equity payback; project payback	Year 6 (2034/35); Year 8 (2036/37)
Fee per accredited MW-day; share of the customer's rent	$1,010; 26.6%

The fee is a floor, proving the investment works rather than a customer will pay for it. The tax credit returns 27% of CAPEX in Year 1, which is why equity pays back in Year 6, and losing its wage or content condition would raise the fee to $33.30 or $34.71, both within the developer's control through its contracts. With a flat fee, escalating O&M and falling ratings, the EBITDA margin falls from 67% to 54% and coverage from 2.41x to 1.42x, so the last year sets the debt.

7. Sensitivity Analysis and Conditions for Success

Every fee below is computed live on the workbook's Sensitivity sheet.

7.1 Inputs and Ranges

Three uncertain inputs are varied one at a time over published ranges, re-solving the fee each time for a 12% equity return, while duration, a design choice, is not varied. Installed cost runs between NREL's 2028 low and high projections, 0.69 and 1.08 of the 2024 cost. The capacity price runs from the 2029/30 floor of $181.91/MW-day to $554.72, PJM's simulation of the 2028/29 auction without its collar. Fixed O&M runs from NREL's literal 4-hour definition, 1.92% of this installed cost, up to the base 4%, which NREL calls the high end, so its range is one-sided.

7.2 Results

Figure 7.1 shows the re-solved fee at each end of each range. Installed cost dominates, at $20.22 to $35.62, and every 10% of CAPEX moves the fee by $3.47; the capacity price gives $31.17 to $22.87, and fixed O&M can only lower the fee, to $22.07. No tested range reaches the $6.10–$15.58 the customer would pay through RBP.

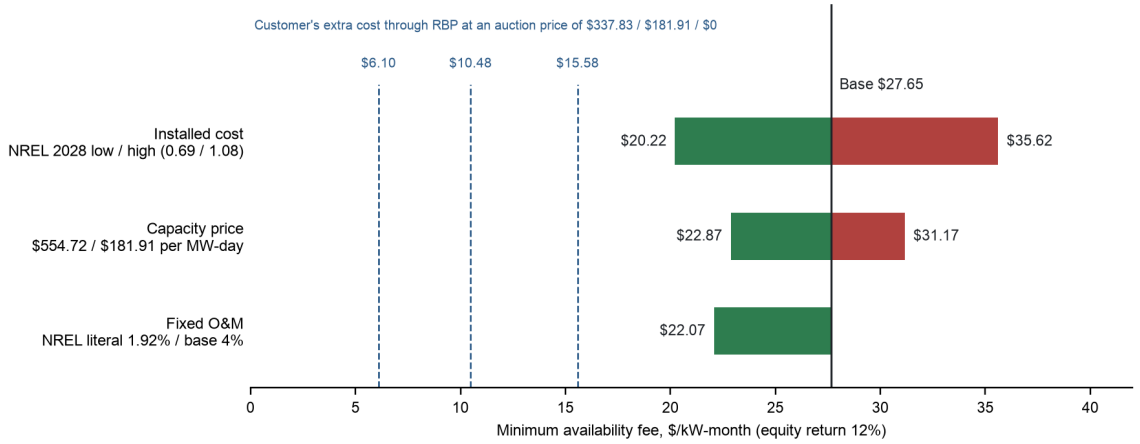


Figure 7.1 Minimum Fee over Each Tested Range Against the Customer's Cost Through RBP

7.3 Thresholds That Change the Conclusion

Once RBP capacity is in service, the customer's ceiling is $6.10–$15.58. Solving for the CAPEX at which the fee meets each ceiling, the RBP route needs $88.2M at $15.58, $68.7M at $10.48 and $52.1M at $6.10, which is 39–65% of the base and below NREL's 2028 low projection ($105.9M), so the next round tests lower-cost designs such as 5 hours ($76.6M, $12.91).

Four further thresholds are yes-or-no. If FERC rejects IRAS, the need disappears and the project is set aside; a 76% storage rating instead of 90% leaves 38.0 MW, 82% of the obligation, and calls for a redesign; missed tax-credit conditions raise the fee to $33.30 or $34.71; and no recharge room raises it to $28.01.

7.4 Conditions for Success

The developer controls CAPEX at or below $135.1M, tax-credit compliance, the schedule and availability; the customer controls the 15-year contract at or above $27.65, the recharge room and the 1.16 MW; PJM and FERC control the 90% rating and IRAS; FERC and the Virginia SCC shape the

customer's alternatives; and Loudoun County approves the land use. The inputs the developer controls keep the fee within $20.22–$35.62, and the base case is financeable at $27.65, so whether a customer signs depends on the conditions that others control.

8. Environmental and Social Considerations

This section is a screening rather than an environmental and social impact assessment (ESIA). It names the likely impact pathways, the evidence behind each, what needs independent study and what an authority must review, and it states no approval outcome.

8.1 Approach and Site Basis

Each issue is traced from project activity to the affected people or resource, the plausible effect, the management action and the consequence for the project, using county and state rules and the published reviews of two comparable batteries. The first is Kifer in Santa Clara, California, 50 MW / 200 MWh at a city substation that also feeds a substation dedicated to one data-center customer, owned and operated by a third party [46], and the second is the battery of up to 1,200 MWh at Biesjesvlei, South Africa [47].

Since 18 March 2025 a new Loudoun data center needs a special exception, while storage above 600 kWh is permitted by right in the industrial districts, so the screening assumes an industrial campus. Under §4.07.02.B of the zoning ordinance, storage there needs at least 5 acres, 100-foot setbacks, no historic resource within 1 mile, decommission and emergency plans, and UL 9540 listing. Scaled by energy, Kifer's site of about 1.2 acres for 200 MWh gives 3.0 acres, and Biesjesvlei's units on about 6 ha for 1,200 MWh give 6.2 acres, or 8.0 acres with laydown and an O&M building, so the battery needs 3.0–8.0 acres.

8.2 Material Impact Pathways

Table 8.1 traces six issues from project activity to their consequence for the project.

Table 8.1 Screening of Material Environmental and Social Issues

Issue	Pathway and plausible effect	Assessment or management	Project consequence
Fire and toxic smoke	500 MWh of cells beside occupied data halls; a worst-case container fire is significant within 10 m, and the 2025 Moss Landing fire damaged about 55% of 100,000 batteries [48]	Hazard analysis on UL 9540A data and fire-authority approval, as at Kifer; emergency plan under ZO §4.07.02.B.10	Layout and spacing; added CAPEX; customer and insurer acceptance
Noise	Inverters, transformers and cooling run day and night, and 24% of Loudoun data-center sites lie within 200 ft of residential zoning; steady, tonal noise is a common complaint [49]	Sound study against the county's 55 dBA limit at residential property lines; enclosures or walls	Equipment placement; barrier cost
Land use and historic resources	A 5-acre storage use where data-center growth is contested	Historic-register screen within 1 mile; pre-application meeting	A historic resource rules the site out
Construction and stormwater	Grading 3.0–8.0 acres causes erosion, runoff and delivery traffic	Construction General Permit for 1 acre or more [50]; traffic plan	Permit needed before grading
Decommissioning	Removing and recycling the cells after 15 years or a fire	Decommission plan; removal within 365 days	Cost not in the model; lender scrutiny
Air quality (positive)	The battery replaces diesel hours in emergencies	None required	A benefit, not quantified

8.3 Effects, Further Study and Authority Review

The setbacks, 5-acre minimum and historic-resource rule decide whether a campus can host the battery; fire spacing and noise decide where. Fire barriers, sound walls, stormwater works and decommissioning security are not in the model. The hazard analysis, sound study, historic screen, stormwater plan and decommissioning plan must be done independently before the county's site-

plan review and DEQ's storage permit by rule [51, § 130], and lenders and insurers will ask for them. Each review must cover the battery as built. None reviews has begun, and no outcome is assumed.

9. Regulatory and Contractual Requirements

Approvals are decisions by public bodies or PJM under law or tariff; contracts are agreements the developer negotiates.

9.1 Approvals and Regulatory Requirements

Table 9.1 follows the Kifer battery's permit table, adding the legal basis, timing and status of each approval.

Table 9.1 Approvals and Regulatory Requirements

Authority	Approval	Legal basis	Timing and status
FERC	IRAS and its BYONC rules	FPA §205	Requested for 12 Oct 2026; pending as of 8 Oct 2026, with 23 protests or limited protests
FERC	RBP timing and cost allocation	FPA §205, §206	Suspended to 28 Feb 2027; hearing on zone allocation, transmission-owner exit and collateral
PJM	Demand-resource registration through a CSP	Manuals 11 and 18 [52]	By 31 May 2029; not started
PJM	Clear a 2029/30 incremental auction	Tariff Att. DD [53]	19 Jul 2028 or 21 Feb 2029; dates tentative
Dominion, PJM	BYONC report	Att. DD-4 §VII(c)	30 days before the auction's parameters are posted; needs IRAS as filed
Dominion	Parallel operation and relays	Schedule GS-4	Before energisation; GS-5 terms to confirm
Loudoun County	Site plan with decommission and emergency plans	ZO §4.07.02	40-day review and 30-day resubmission [54]; not applied
Loudoun County	Building and fire permits	13VAC5-63; 13VAC5-51	Needs UL 9540A data; NFPA 855 edition to confirm
Loudoun County, DEQ	Stormwater coverage	9VAC25-875	Required before grading
Virginia DEQ	Storage permit by rule	9VAC15-100	After the site plan; no review period
Virginia SCC	Non-utility storage permit, 1 MW or more	20VAC5-335	No decision deadline [55, § 80]
IRS	Tax credit and its transfer	26 U.S.C. §48E, §6418	2028 start; election with the 2029 return

9.2 Principal Contracts

Table 9.2 lists the agreements the project must sign, their key terms and how each allocates risk.

Table 9.2 Principal Contracts

Agreement	Counterparty	Key terms and risk allocation	Timing
Availability service agreement	Customer	15-year take-or-pay at ≥ $27.65, owed even if IRAS fails; customer arranges 1.16 MW, keeps recharge room and gives credit support; developer bears PJM charges	Term sheet at the gate; signed before close
Site lease and easements	Campus owner	≥ 5 acres for 15 years plus decommissioning; pad provided in kind	Before the site plan
Battery supply and EPC	Integrator, EPC contractor	Fixed price ≤ $135.1M; delay damages tied to PJM dates; performance guarantees; UL 9540; wage and FEOC records	Before close

Agreement	Counterparty	Key terms and risk allocation	Timing
Long-term service	Integrator or O&M provider	Outages ≤ 8.94%; augmentation within the 4% O&M	With the EPC
CSP agreement	PJM member	Registration, offers, settlement and liability for charges	Before the auction
Loan and tax-credit transfer	Lenders; credit buyer	$55.7M at 7.02% and 1.40x DSCR; $40.5M credit sold for $36.5M	Financial close, end of 2027
Insurance	Insurers	Fire, damage and liability at 0.4% of CAPEX	Before construction

9.3 Dependencies and Open Confirmations

The FERC order comes first, the term sheet and lease then define the site, and the vendor's layout and UL 9540A data feed the site plan and fire permits, which precede DEQ's permit by rule. The PJM path runs in parallel, and financial close needs them all. PJM must still confirm the 90% rating and the all-hours rule, Dominion the terms of its Schedule GS-5 tariff, DEQ the interconnection document for a non-exporting battery, the SCC its review time, and Loudoun the campus's district and historic resources.

10. Risk Analysis and Development Plan

Each risk is traced from the uncertainty to its consequence, the party that controls or bears it, the response and the residual exposure, and priority follows likelihood times consequence. A consequence counts as high when the project would be set aside, a delivery year would be lost or the minimum fee would leave the tested range of $20.22–$35.62.

10.1 Priority Risks

Table 10.1 groups eight risks by priority, and each response names the development task, T1 to T12, or the contract term that carries it.

Table 10.1 Priority Risks

Risk	Consequence	Borne by	Response	Residual exposure
Priority: high consequence; likelihood high or unresolved before the gate				
Customer will not sign at ≥ $27.65 for 15 years, as RBP may later cost it $6.10–$15.58	No contracted revenue, no debt	Customer; developer sets the offer	Customer-side valuation and term sheet before the gate (T6); price 5-hour, low-cost and shorter-term options	RBP allocation and Virginia rules
FERC rejects or changes IRAS, or IRAS lapses	Before signing, set aside; after, a weaker incentive to pay	FERC; customer after signing	Limited spend until FERC's order (T1); take-or-pay with credit support	Lapse risk over the term
PJM rejects the 90% rating or tightens the all-hours rule	At 76%, 38.0 MW, and the customer needs 8.2 MW more	PJM	PJM's written confirmation before the gate (T2)	Ratings fall to 75%, as modelled
Monitor: high consequence, lower likelihood				
Not in service and registered by 1 June 2029	The 2029/30 year is lost	Developer; SCC, county, vendor	Early SCC filing (T3); EPC delay damages (T11); 2030/31 fallback	One to five months of float
Campus cannot host the battery	Relocation or extra cost	Developer, customer	Site screen before the gate (T8); sound study (T9)	Local opposition
Battery fire	Damage, outage, lost acceptance	Developer, vendor, insurers	UL 9540A data (T10); hazard analysis and emergency plan (T9)	Low likelihood, high consequence
Manage: consequence within the tested fee range				
CAPEX above base or tax-credit conditions missed	Fee up to $35.62, $33.30 or $34.71	Developer	Competitive RFP (T10); fixed-price EPC (T11)	Cell tariffs

Risk	Consequence	Borne by	Response	Residual exposure
No overnight recharge room	Fee $28.01	Customer	Recharge clause in the service agreement (T7)	GS-5 terms unconfirmed

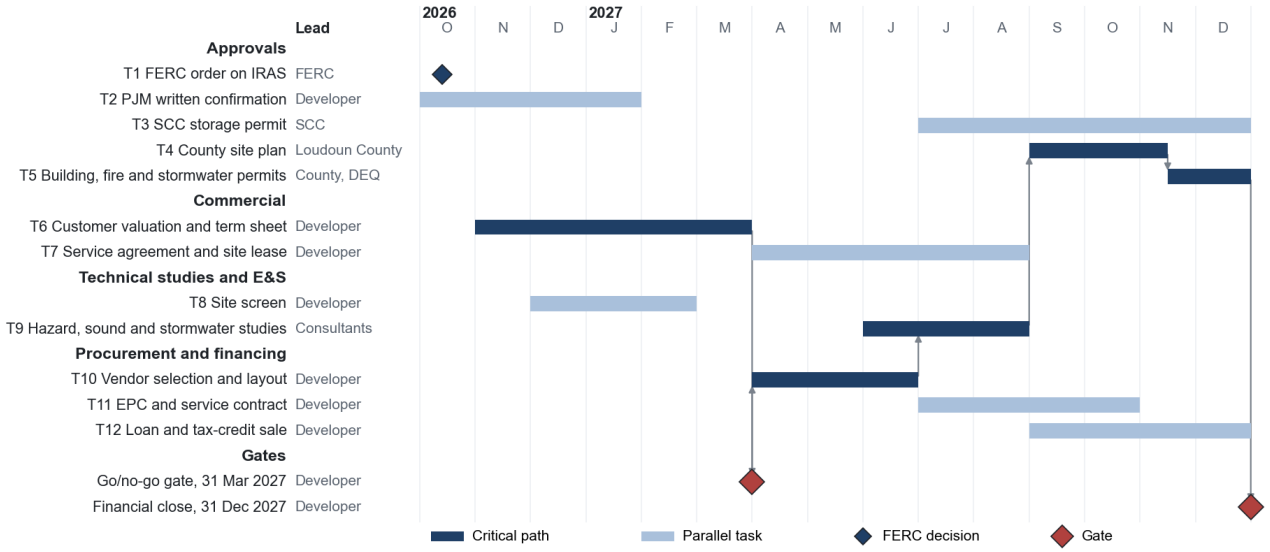


Figure 10.1 Development Schedule from October 2026 to Financial Close on 31 December 2027

10.2 Development Plan

The go/no-go gate on 31 March 2027 comes after the five external decisions that can be known in time. FERC's order on IRAS is requested for 12 October 2026, PJM may refile RBP by 29 October 2026, the 2029/30 auction result is due on 22 December 2026, Virginia's interruptible tariffs by 1 February 2027, and RBP can take effect on 28 February 2027 at the earliest. Before the gate the developer spends only on PJM's written confirmation, the customer-side valuation and term sheet, and the site screen. FERC decides T1; the developer leads T2, T6–T8 and T10–T12, with PJM, the customer, the integrator and lenders; consultants prepare the studies (T9); and the SCC, Loudoun County and DEQ decide T3–T5. Figure 10.1 shows the tasks by workstream and marks the critical path, and Table B.1 gives each task's timing, dependencies and duration basis.

10.3 Critical Path, Float and Responses

The critical path runs from FERC's order through the valuation and term sheet, the gate, the vendor layout, the studies, the county site plan and the permits to financial close and construction. A 12-month build, Lazard's benchmark for utility solar-plus-storage, leaves five months of float, and a 16-month build like Kifer's leaves one; the SCC permit, with no decision deadline, is the open item, and if close slips past spring 2028 the project targets 2030/31. The customer, IRAS and rating risks are tested before the gate, the schedule risk is managed by the early SCC filing and the fallbacks, the site and fire risks by the site screen and studies, and the cost and recharge risks through the EPC and the service agreement.

11. Final Appraisal and Next Steps

11.1 Recommendation

Of the four options, studying further fits the evidence best. Proceeding now would commit spend before FERC rules on IRAS and before any customer has accepted the price; redesigning now would be premature, since the 5-hour and shorter-term options have not been put to customers; and setting the project aside would ignore a real shortfall and a financeable base case.

11.2 Evidence and Assumptions

The case for continuing rests on the 6,831 MW shortfall and the recent dispatch record, a design that accredits 97.5% of the obligation, a base case financeable at $27.65, and the absence of any cheaper priced route by 2029/30. What remains to be shown is the price once RBP capacity is in service, $6.10–$15.58 against $27.65, and whether a customer accepts a 15-year term. Beyond the five conditions for success, the judgment also assumes capacity prices at or above the $181.91 floor.

11.3 Remaining Uncertainties

FERC's order on IRAS, the 2029/30 auction price, RBP's effective date, PJM's confirmation and the customer's willingness to pay are all known by the gate, and RBP's zone allocation probably is. Two are not: Virginia's allocation of RBP costs, due as a model on 1 January 2028, is the customer's risk under take-or-pay, and the SCC's review time is carried as schedule float.

11.4 Next Steps and Decision Rules

Before committing further resources, the developer should value the service from the customer's side, setting the expected reduction hours, tenants' service credits and outage costs, and the rent gap between firm and interruptible capacity against each alternative route. It should then price the options this points to, namely 5 hours ($12.91), NREL's low cost ($20.22) and a term matched to the years before RBP capacity arrives, update the market case as FERC and PJM decide, obtain PJM's written confirmation by January 2027 and screen campuses by February 2027.

At the gate, the project proceeds if FERC has accepted IRAS, PJM has confirmed the rating, a term sheet meets the minimum fee and a campus has passed the screen. It is redesigned if customers accept only a lower price or a shorter term, and set aside if FERC rejects IRAS or no customer accepts any priced option.

References

[1] PJM Interconnection. (2026, July 14). 2028/2029 base residual auction report. PJM Interconnection, L.L.C.

[2] PJM Interconnection. (2026, July 27). CIFP framework for service during periods of insufficient resource adequacy – PJM proposal [Board decision, executive summary]. PJM Interconnection, L.L.C.

[3] PJM Interconnection. (2026, August 13). Proposal of PJM Interconnection, L.L.C. to establish Interim Resource Adequacy Service and a Large Load Registry to address system reliability, Docket No. ER26-3515-000. Federal Energy Regulatory Commission.

[4] Cole, W., Ramasamy, V., & Turan, M. (2025, June). Cost projections for utility-scale battery storage: 2025 update (NREL/TP-6A40-93281). National Renewable Energy Laboratory.

[5] PJM Interconnection. (2026). Emergency procedures postings, January 2017 – September 2026 [Data set]. Retrieved September 26, 2026, from https://emergencyprocedures.pjm.com

[6] PJM Interconnection. (2026). ELCC class ratings for the 2029/2030 base residual auction. PJM Interconnection, L.L.C.

[7] PJM Interconnection. (2026, September 3). 2029/2030 RPM base residual auction planning period parameters. PJM Interconnection, L.L.C.

[8] Joint Legislative Audit and Review Commission. (2024, December 9). Data centers in Virginia (Report No. 598). Commonwealth of Virginia.

[9] CBRE. (2026). North America data center trends H1 2026. CBRE Group, Inc.

[10] Donnellan, D., Lawrence, A., Bizo, D., Rogers, O., Judge, P., Davis, J., Smolaks, M., & Weinschenk, R. (2025, July). Uptime Institute global data center survey 2025 (UII Keynote Report 180). Uptime Institute.

[11] How do I demonstrate continuous compliance with the emission limitations, operating limitations, and other requirements?, 40 C.F.R. § 63.6640(f) (2022).

[12] Permits for data centers; emission limits for certain engine-generator sets, Va. Code Ann. § 10.1-1322.6.

[13] PJM Interconnection, L.L.C., 195 FERC ¶ 61,253 (2026) (order in Docket No. ER26-2319-000, June 26, 2026).

[14] Uptime Institute. (2026, May 13). Uptime announces annual outage analysis report 2026 [Press release]. https://uptimeinstitute.com/about-ui/press-releases/uptime-announces-annual-outage-analysis-report-2026

[15] Dominion Energy Virginia. (2025, September). 2025 20-year data center forecast [Presentation to the Virginia State Corporation Commission, September 16, 2025].

[16] PJM Interconnection. (2026, July 31). Reliability backstop procurement to address resource adequacy concerns stemming from large loads, Docket No. ER26-3380-000. Federal Energy Regulatory Commission.

[17] Lazard. (2026, July). Levelized cost of energy+ (LCOE version 19.0; LCOS version 11.0). Lazard Ltd.

[18] Federal Energy Regulatory Commission. (2026, September 29). Order accepting Reliability Backstop Procurement and establishing further procedures, PJM Interconnection, L.L.C., Docket Nos. ER26-3380-000 and EL26-108-000 (eLibrary accession no. 20260929-3099).

[19] Enchanted Rock, Inc. (2026). Draft registration statement (Form DRS). U.S. Securities and Exchange Commission.

[20] Generation of electricity from renewable and zero carbon sources, Va. Code Ann. § 56-585.5(E) (2026) (Virginia Clean Economy Act).

[21] Monitoring Analytics, LLC. (2026, September 3). Protest of the Independent Market Monitor for PJM, Docket No. ER26-3515-000. Federal Energy Regulatory Commission.

[22] PJM Interconnection. (2026, September 23). Answer to protests and comments, Docket No. ER26-3515-000. Federal Energy Regulatory Commission.

[23] Loudoun County. (2025, March 18). Data centers: Land use considerations. Retrieved October 3, 2026, from https://www.loudoun.gov/6410/Land-Use-Considerations

[24] PJM Interconnection. (2026, September 24). PJM Manual 18: PJM capacity market (Rev. 63). PJM Interconnection, L.L.C.

[25] PJM Interconnection. (2026, August 31). Preliminary ELCC class ratings for delivery years 2028/2029–2042/2043 [Prepared for the Reliability Backstop Procurement]. PJM Interconnection, L.L.C.

[26] California Public Utilities Commission, Energy Division. (2025, November 3). Workshop on unforced capacity methodology and preliminary results [Workshop slides, Rulemaking 23-10-011].

[27] Internal Revenue Service. (2026, January 5). Clean electricity investment credit. U.S. Department of the Treasury.

[28] Norton Rose Fulbright. (2026, January). Cost of capital: 2026 outlook. Project Finance NewsWire.

[29] Board of Governors of the Federal Reserve System. (2026, September 4). Selected interest rates (daily) — H.15 [Data release; values for September 3, 2026].
[30] Norton Rose Fulbright. (2025, January). Cost of capital: 2025 outlook. Project Finance NewsWire.
[31] Virginia Electric and Power Company. (2025, December 9). Schedule GS-4: Large general service, primary voltage [Tariff filing, effective January 1, 2026]. Dominion Energy.
[32] PJM Interconnection. (2023, July 17). Winter Storm Elliott event analysis and recommendation report. PJM Interconnection, L.L.C.
[33] Loudoun County. (n.d.). Loudoun County zoning ordinance: Chapter 3, Uses; Chapter 4, Use-specific standards; Chapter 12, Definitions. Retrieved October 3, 2026, from https://www.loudoun.gov/DocumentCenter/View/176097, /176098 and /176106
[34] Internal Revenue Service. (2026, February 12). Notice 2026-15: Material assistance cost ratio and prohibited foreign entity rules for clean energy credits. U.S. Department of the Treasury.
[35] PJM Interconnection, L.L.C., 195 FERC ¶ 61,076 (2026) (order in Docket No. ER26-1556-000, April 28, 2026).
[36] PJM Interconnection. (2025, July 22). 2026/2027 RPM base residual auction report. PJM Interconnection, L.L.C.
[37] PJM Interconnection. (2025, December 17). 2027/2028 base residual auction report. PJM Interconnection, L.L.C.
[38] Pacific Northwest National Laboratory. (2024, April). Energy storage cost and performance assessment: LCOS workbook v.2024 documentation. U.S. Department of Energy.
[39] PJM Interconnection. (2026). Data Miner 2: Hourly load, metered (hrl_load_metered) and day-ahead hourly LMPs [Data set]. Retrieved September 3, 2026.
[40] Transfer of certain credits, 26 U.S.C. § 6418.
[41] Other special rules, 26 U.S.C. § 50(c)(3).
[42] Internal Revenue Service. (2025). Publication 946: How to depreciate property. U.S. Department of the Treasury.
[43] Tax imposed, 26 U.S.C. § 11(b).
[44] Imposition of tax, Va. Code Ann. § 58.1-400.
[45] Net operating loss deduction, 26 U.S.C. § 172(a).
[46] City of Santa Clara. (2023, July). Final mitigated negative declaration and supporting initial study for Silicon Valley Power's Kifer Receiving Station BESS project. City of Santa Clara, California.
[47] Council for Scientific and Industrial Research. (2024, September). Environmental impact assessment report: Proposed development of a solar photovoltaic facility (Biesjesvlei PV1), battery energy storage system (Biesjesvlei BESS 1) and 132 kV overhead power line near Smithfield, Free State [Prepared for Scatec Africa (Pty) Ltd and Veroniva (Pty) Ltd]. CSIR.
[48] U.S. Environmental Protection Agency. (n.d.). Moss Landing Vistra battery fire response. Retrieved October 3, 2026, from https://www.epa.gov/ca/moss-landing-vistra-battery-fire
[49] Loudoun County. (n.d.). Noise & air quality concerns. Retrieved October 3, 2026, from https://www.loudoun.gov/6405/Noise-Air-Quality-Concerns
[50] Erosion and Stormwater Management Regulation, 9 Va. Admin. Code 25-875.
[51] Small Energy Storage Facilities Permit by Rule, 9 Va. Admin. Code 15-100.
[52] PJM Interconnection. (2026, July 28). PJM Manual 11: Energy & ancillary services market operations (Rev. 137). PJM Interconnection, L.L.C.
[53] PJM Interconnection. (2026). RPM auction schedule [Data set]. PJM Interconnection, L.L.C.
[54] Designated agent to act on proposed final plat, Va. Code Ann. § 15.2-2259.
[55] Regulations Governing the Deployment of Energy Storage, 20 Va. Admin. Code 5-335.

Appendix A. Recharge Between Consecutive Dispatch Days

Table A.1 Recharge Needed Between Consecutive Dispatch Days

Pair of days	Hours between events	Power above load to refill fully (MW)	Power to cover the next day's event (MW)
23–24 Dec 2022	7.75	32.3	32.3
23–24 Jun 2025	17.0	24.2	13.8
24–25 Jun 2025	17.0	24.2	3.5
28–29 Jul 2025	17.0	18.2	6.9
18–19 May 2026	17.0	20.8	12.1
2–3 Jul 2026	14.5	32.5	28.4

Appendix B. Development Plan Detail

Table B.1 Development Tasks

Task	Start → finish	Depends on	Duration basis
T1 FERC order on IRAS	by 12 Oct 2026	—	Requested effective date
T2 PJM written confirmation of registration, rating and "all hours" rule	Oct 2026 – Jan 2027	T1	Estimate: written questions and staff guidance
T3 SCC permit	Jul–Dec 2027 (planned)	T10; site information	No decision deadline
T4 County site plan (first review and one resubmission)	Sep – mid-Nov 2027	T7, T9	40 + 30 days
T5 Building and fire permits, stormwater coverage, DEQ notice of intent	mid-Nov – Dec 2027	T4	Estimate; DEQ's ≤ 10-acre route sets no review period
T6 Customer-side valuation and non-binding term sheet	Nov 2026 – Mar 2027	T1	Estimate: a desk valuation of the customer's alternatives, then the price, term and risk terms of the service agreement, informed by the 2029/30 auction and RBP's effective date
T7 Service agreement and site lease	Apr–Aug 2027	Gate	Estimate
T8 Site screen: zoning district, historic resources within 1 mile, nearest homes	Dec 2026 – Feb 2027	T6 (which campus)	Estimate: desk review of county and state registers
T9 Studies and plans: hazard mitigation, sound, stormwater, emergency operations, decommissioning	Jun–Aug 2027	T10 (layout)	Estimate
T10 Vendor RFP, selection, layout and UL 9540A data	Apr–Jun 2027	Gate	Estimate
T11 EPC and long-term service agreement	Jul–Oct 2027	T10	Estimate
T12 Loan and tax-credit transfer	Sep–Dec 2027	T7, T11	Estimate
Go/no-go gate	31 Mar 2027	T1, T2, T6, T8	—
Financial close	31 Dec 2027	T4, T5, T7, T11, T12	—